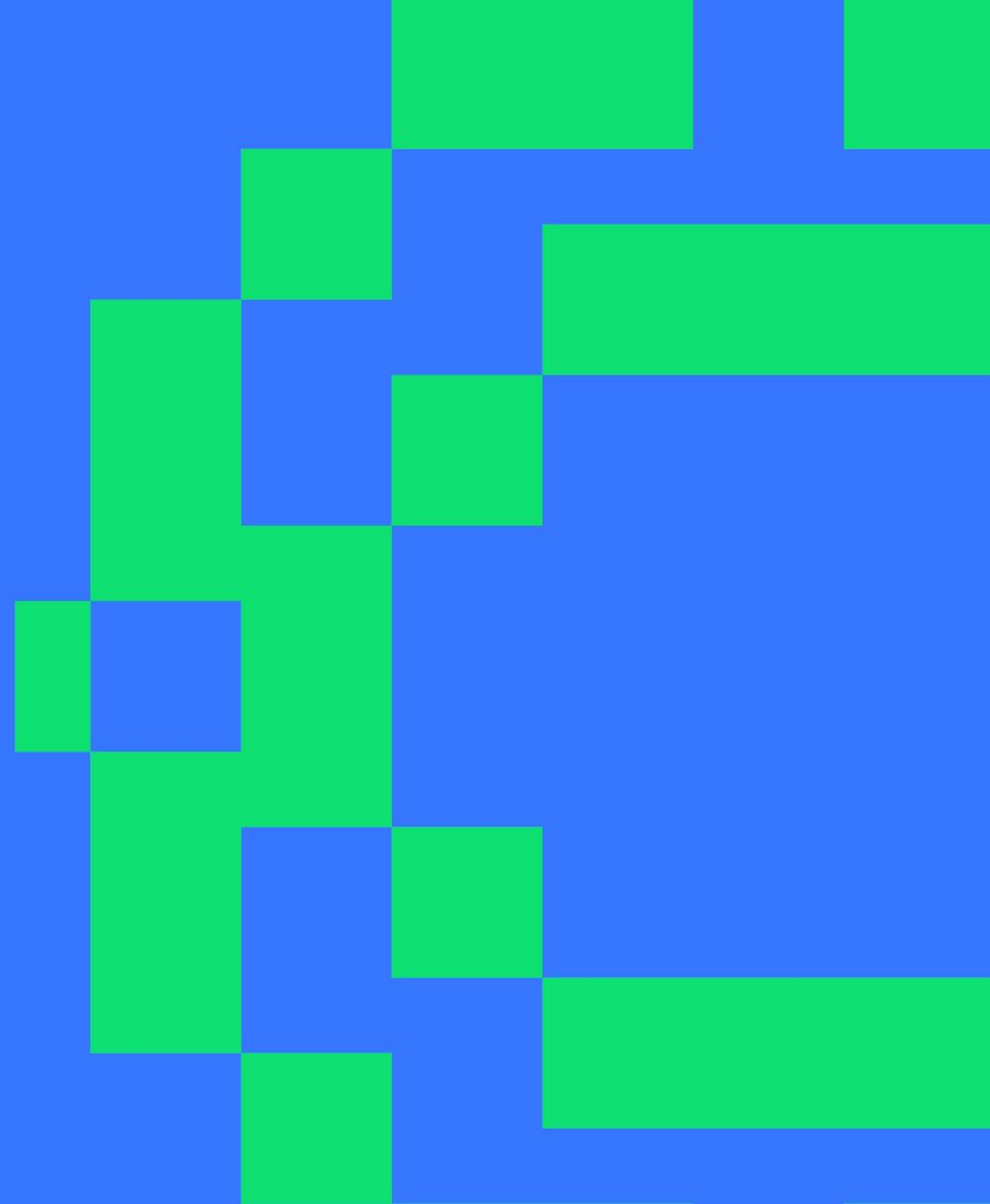


F1 Race Strategy Intelligence Platform

PySpark Big Data Platform · Formula 1 Constructor
Decision-Making

NOVA IMS · Big Data Analytics 2026

5th June 2026



THE CLIENT CHALLENGE

Why Formula 1 is a Big Data Problem. 74 years of race data, millisecond decisions

Formula 1 generates one of the densest operational datasets in sport. Every race weekend produces 50,000+ data points per car. The client needs a platform that joins 74 years of history with live race-day intelligence to make faster, better-informed decisions.

The four dimensions of the challenge:

VOLUME

589K lap-time records
14 tables · 1950–2024

VELOCITY

Pit decisions in seconds
Live telemetry

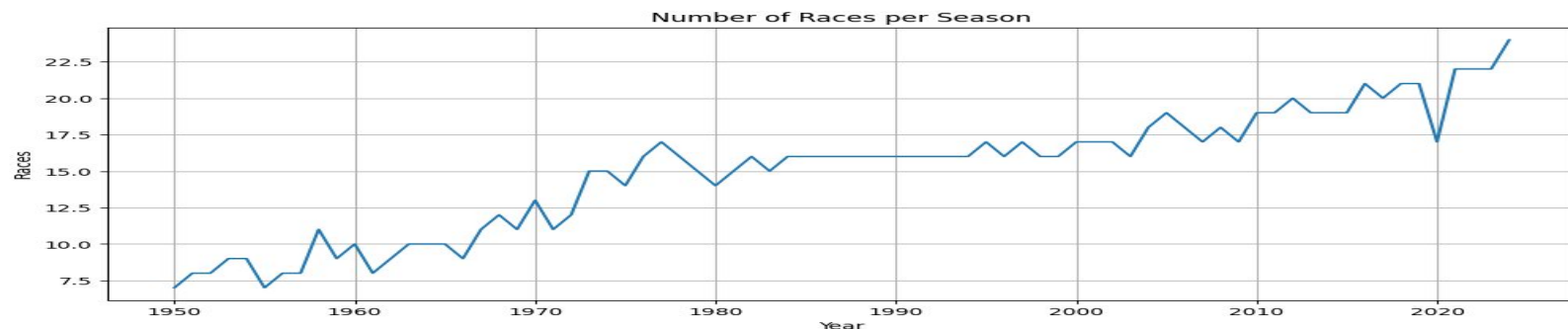
VARIETY

Results, quali, pit stops
standings, laps

VERACITY

Missing data · Status
inconsistencies
Era schema drift

Calendar growth: races per season 1950–2024 (Source: Notebook 2)



TECHNOLOGY CHOICE

Why Apache Spark?

- ▶ **Processes hundreds of thousands of lap records today scales to millions as live telemetry expands**

A single race weekend produces 50,000+ data points per car. Spark distributes this across cores automatically, no manual optimisation required.

- ▶ **One platform: history, ML, and live data**

Spark handles batch analytics, machine learning pipelines, and live streaming in a single system. No tool-switching mid-race.

- ▶ **Scales horizontally as data grows**

As telemetry sensors, weather feeds, and tyre data are added the platform scales out, not up. No re-architecture required.

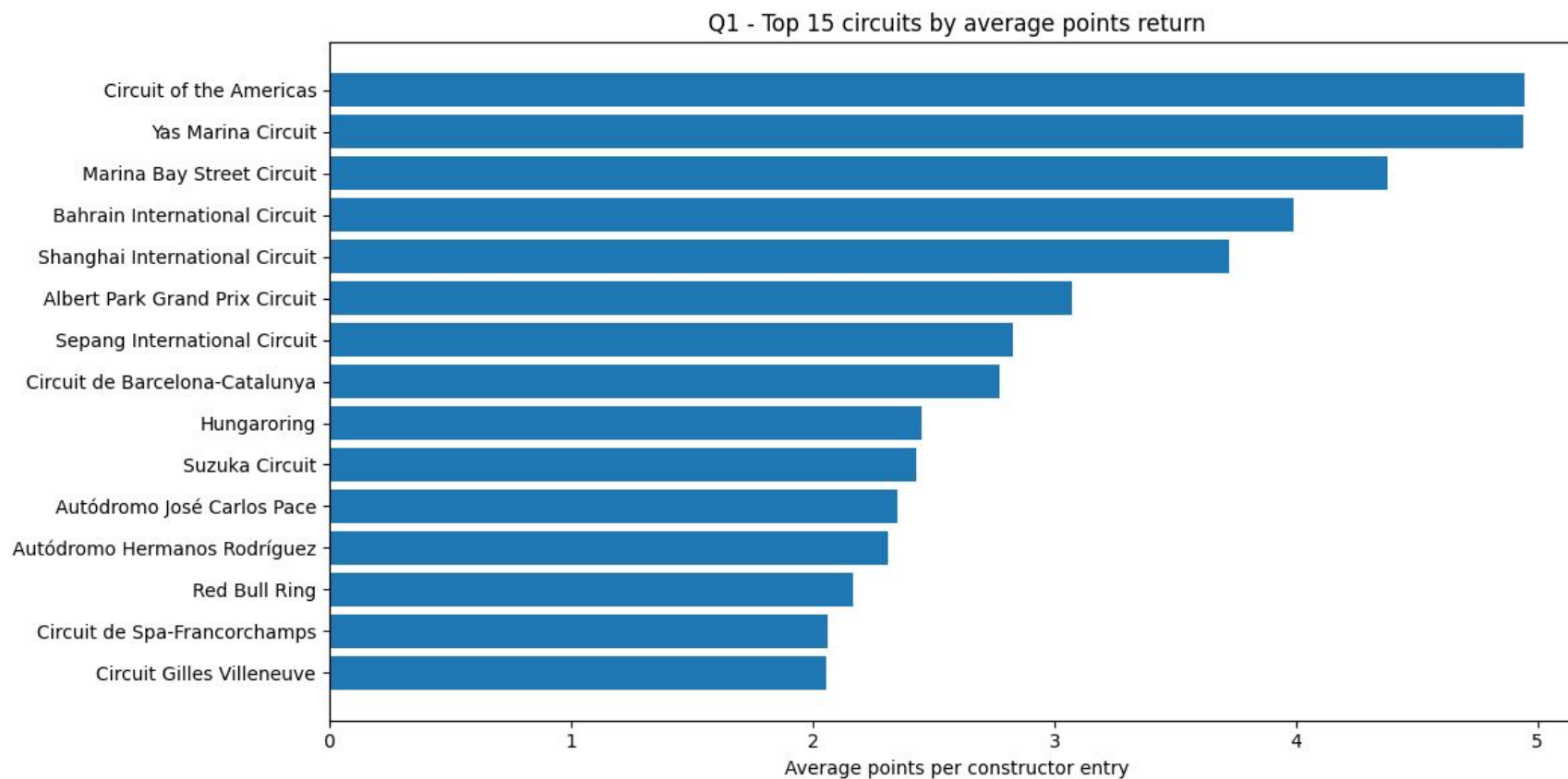
- ▶ **Fault-tolerant by design**

If a node fails mid-calculation Spark re-runs only the affected partition automatically. No data loss, no manual recovery.

Bottom line: Spark is the only open-source engine that does all of this without leaving the platform.

STRATEGIC INSIGHT 1 / 2

Circuit Profitability — avg points return per constructor entry by circuit



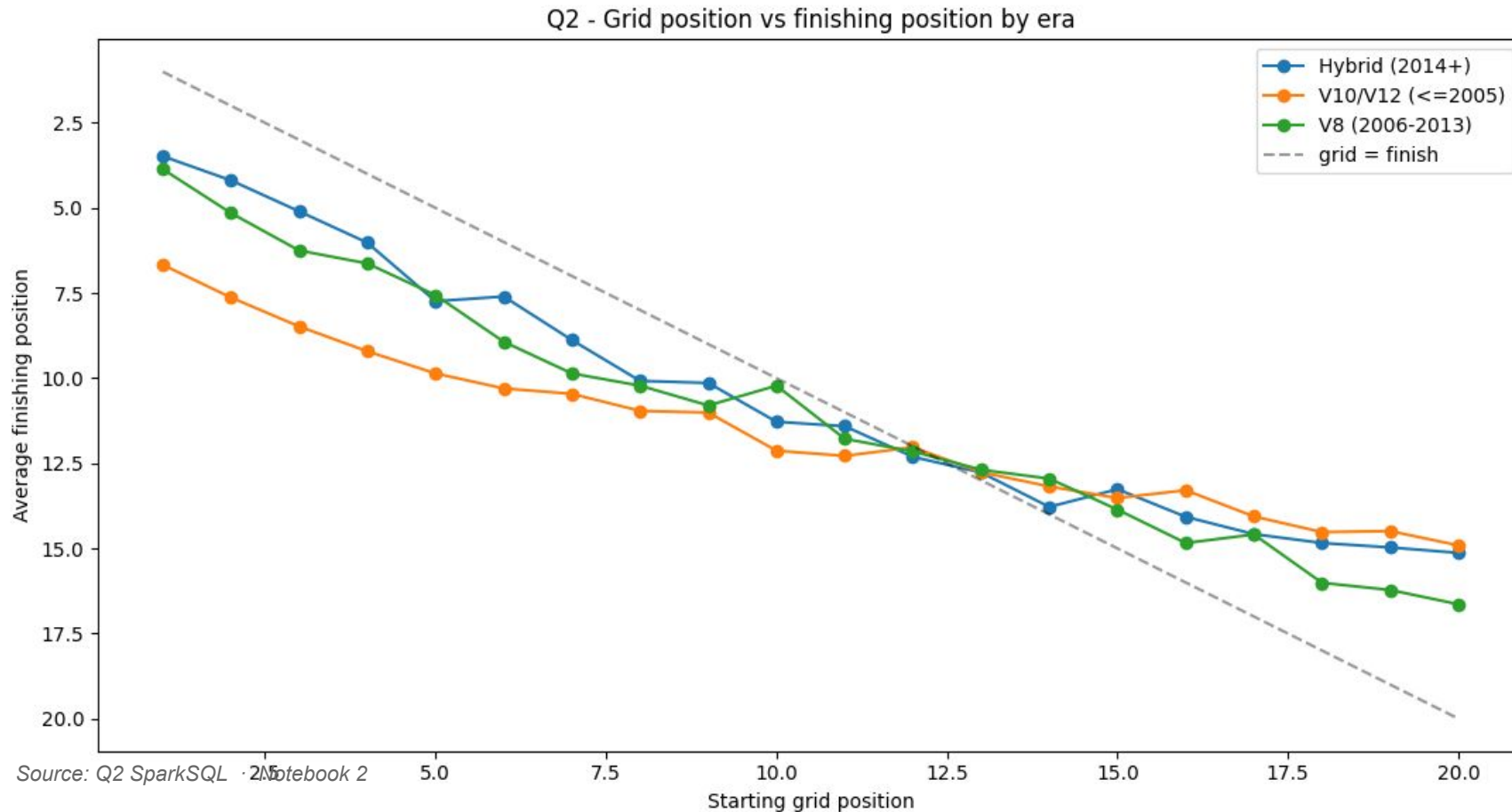
Source: Q1 SparkSQL · Notebook 2

Key Findings

- COTA & Yas Marina top the table (~5.0 pts/entry). Both were added after 2010 when the points system increased from 10 to 25 pts per win, making their average structurally higher.
- Marina Bay (Singapore) ranks 3rd, a street circuit that rewards reliability over outright pace
- All top-5 circuits entered the calendar in 2004 or later, mostly racing under the 25-point era (2010+). Spa-Francorchamps and Gilles Villeneuve, near the bottom, carry decades of older 8-to-10-point data that drags their average down.
- Recommendation: treat the ranking as a points era effect, not circuit quality. The real strategic metric is average points at circuits still on the current calendar under the 25 point system.

STRATEGIC INSIGHT 2 / 2

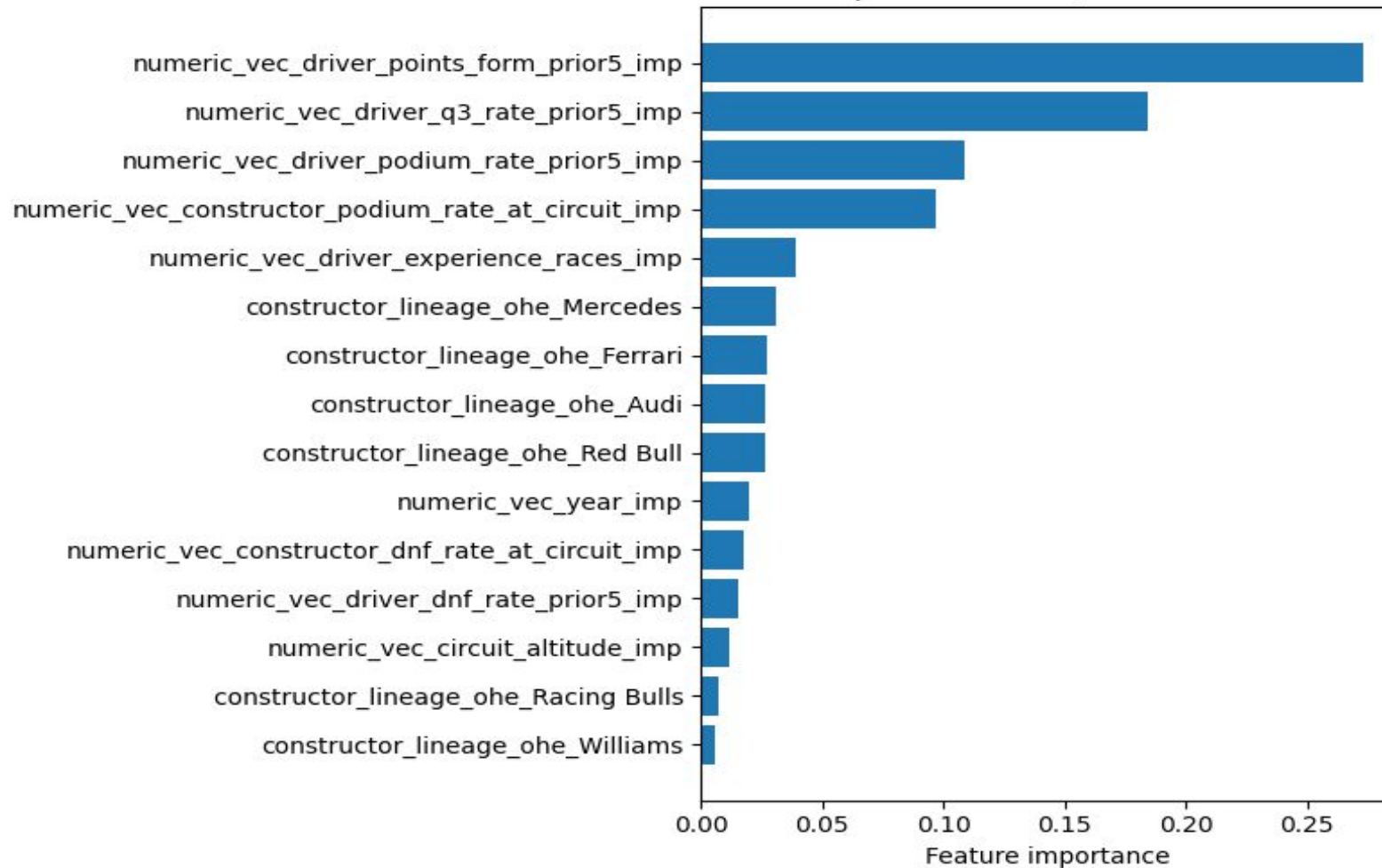
Does Qualifying Actually Matter? — grid position vs finishing position by era



Key Findings

- ▶ A better grid slot means a better average finish in every era — grid 1 has the lowest (best) average finish across all three eras
- ▶ In the hybrid era the line sits closest to the diagonal: where you start predicts where you finish more tightly than before, so modern races shuffle less
- ▶ Part of the bigger early-era gap is reliability, not qualifying — older cars retired far more often, which drags their average finish down regardless of grid
- ▶ Implication: as racing has become more deterministic, qualifying and track position matter more today — prioritise them at circuits where overtaking is hard

Model 1 — Top 15 Features (RandomForestClassifier)



PREDICTION TOOL 1 / 2

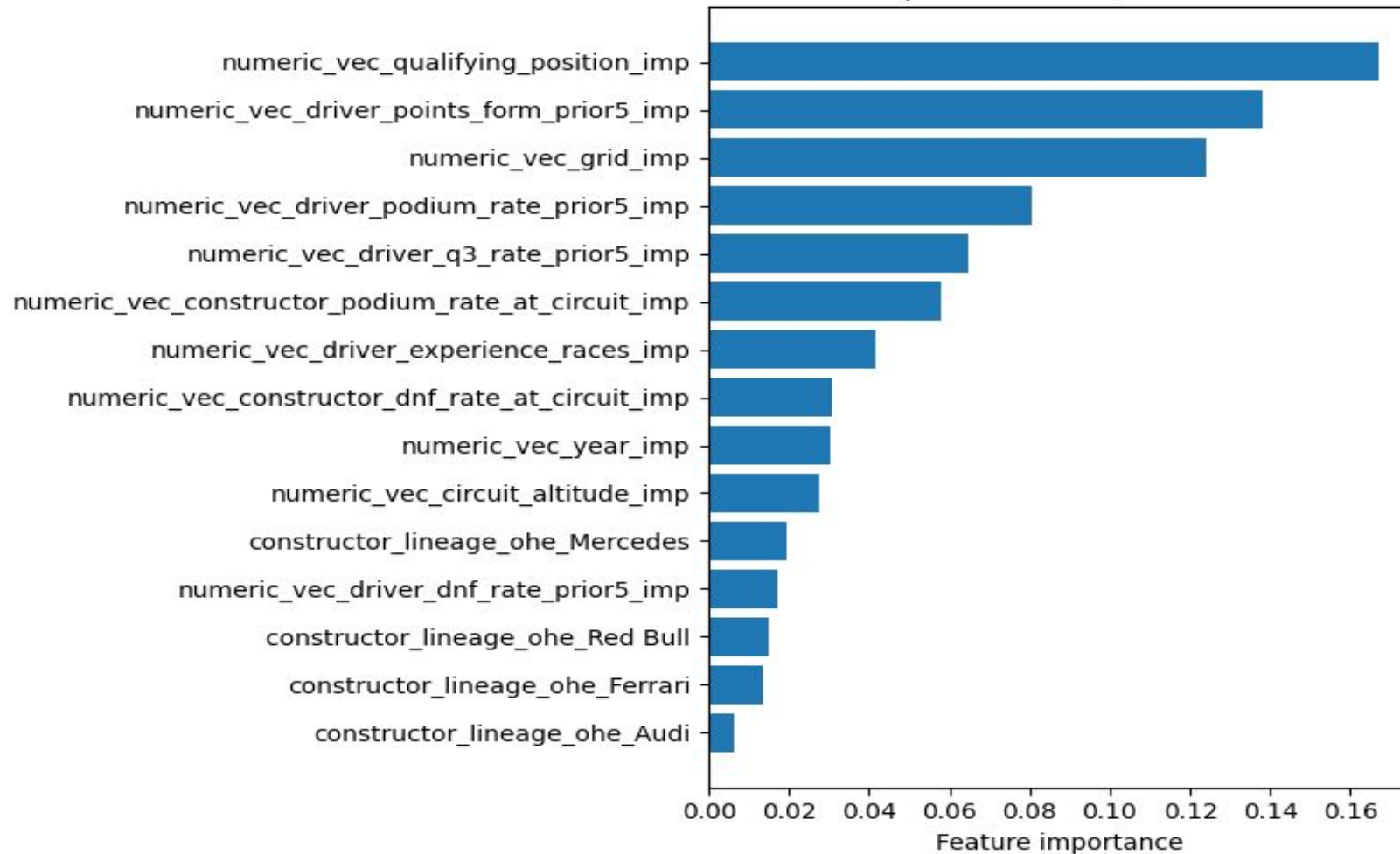
Will We Make Q3?

Qualifying Predictor

- Predicts whether a driver will reach Q3 before the weekend begins
 - 88% ROC-AUC · 78% accuracy · PR-AUC = 0.878 · 5-fold CV
 - Winner: RandomForestClassifier · 80/20 held-out test set (823 rows)
 - Outputs a probability score, the team sets the risk threshold
 - Benchmarked against GBT, MLP, and LogReg
- RF wins on both CV-AUC and test-AUC

Source: Notebook 3 · RandomForestClassifier + LogisticRegression baseline

Model 2 — Top 15 Features (RandomForest-mc)



PREDICTION TOOL 2 / 2

What Will Our Race Outcome Be?

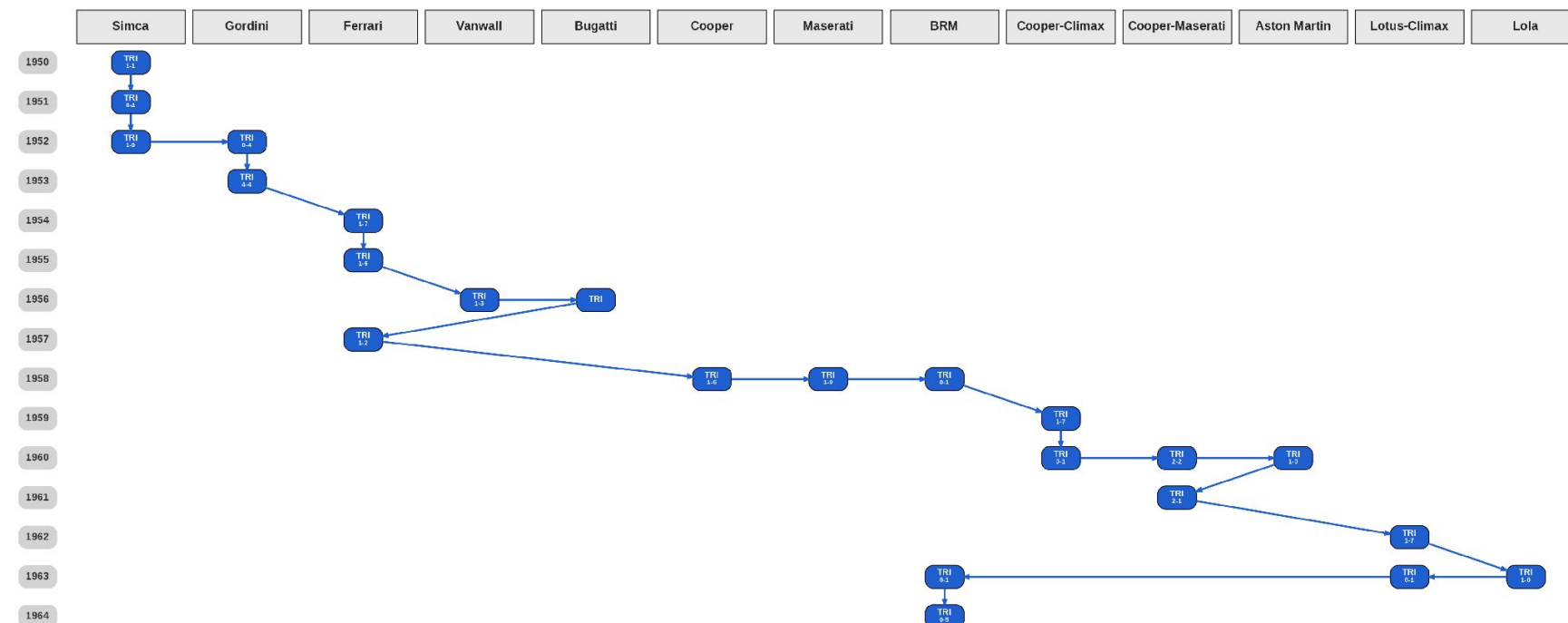
Race Outcome Predictor

- ▶ 5-class prediction: Win · Podium · Points · Finisher-NP · DNF
- ▶ Weighted F1 = 0.51 · Accuracy = 54% · Baseline (majority class) = 20% F1
- ▶ Qualifying position is the single strongest predictor, confirms Saturday's priority
- ▶ Constructor circuit history is second, prior podium rate at a venue is a genuine edge
- ▶ Provides probability distribution across all 5 outcomes, not just a binary flag

Source: Notebook 3 · RandomForest-mc · TrainValidationSplit + held-out test (80/20)

GRAPH ANALYTICS

Why Does Maurice Trintignant Top the PageRank? · 861-driver teammate network · 16,042 edges



Maurice Trintignant · 13 constructors · 1950–1964 (Notebook 2 · `describe_cluster()`)

Source: Notebook 4 · GraphFrames PageRank + LPA + BFS · Notebook 2 · `describe_cluster()`

PageRank 5.70

Highest of all 861 drivers

► 13 constructors over 14 seasons, each team adds entirely new teammates to his edge list

► Multi-team years (1956, 1960, 1963) multiply his connections far beyond single-team drivers

► 2nd: Stirling Moss 5.49 · 3rd: Jack Brabham 5.01, all era-bridging drivers

► Constructor lineage DAG: 32 nodes · 22 edges, Aston Martin highest dynasty PageRank

LIVE RACE DASHBOARD · BONUS

Spark Structured Streaming — Abu Dhabi Grand Prix 2022

LIVE LEADERBOARD | Abu Dhabi 2022 | Lap 1

P	COD	DRIVER	LAP TIME	GAP / ΔPOS
1	VER	Max Verstappen	1:32.198	LEADER
2	PER	Sergio Pérez	1:33.251	+1.053s
3	LEC	Charles Leclerc	1:34.192	+1.994s
4	HAM	Lewis Hamilton	1:34.864	+2.666s ▲ 1
5	SAI	Carlos Sainz	1:35.436	+3.238s ▼ 1
6	NOR	Lando Norris	1:35.908	+3.710s ▲ 1
7	RUS	George Russell	1:36.506	+4.308s ▼ 1
8	OCO	Esteban Ocon	1:36.964	+4.766s

How it works

- ▶ readStream with maxFilesPerTrigger=1, lap files dropped every 2 s
- ▶ withWatermark (10 s) handles late-arriving telemetry without blocking
- ▶ foreachBatch delivers each micro-batch to the live leaderboard handler
- ▶ Position changes (▲ ▼) computed in real time from prior-lap window
- ▶ Audit log written to Parquet in append mode, immutable race record
- ▶ Validated on Abu Dhabi 2022 · fastest lap 89,968 ms · 20 drivers

Source: Notebook 5 · Spark Structured Streaming · actual notebook output shown

Thank You!

Any questions?

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Acreditações e Certificações da NOVA IMS



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